농산물 신선 물류 공급망 모니터링을 위한 다이나믹 QR 코드 및 디지털 트윈

기술 통합 연구

Integration of Dynamic QR Code and Digital Twin Technology for Agricultural Fresh Product Supply Chain Monitoring

한성민, 이효은, 이동수, 김성민*

전북대학교 스마트농생명시스템학과 (Jeonbuk National University)

JBNU LOGO

Abstract

The fresh agricultural product supply chain involves complex and multi-stage processes from production to consumption.
However, fragmented

Result and Discussior

1. 시스템 구성 및 아키텍처 (Systei Architecture)

information sharing has long hindered effective quality control and the stable management of supply and demand. To address these challenges, this study proposes an advanced platform for intelligent end-to-end logistics management by integrating dynamic QR code-based data collection with digital twin technology, which mirrors real-time field data in a virtual environment.

Figure 1. 통합 디지털 트윈 공급망의 개념적 프레임워크.

2. 기술 스택 및 동기화 엔진 (Technical Stack)

- 통신 **MQTT 프로토콜:** 실시간 양방향 데이터 동기화 구현.
- 저장 **Littlefs 파일 시스템:** 임베디드

The proposed system connects data collected via dynamic QR codes across farms, agricultural product processing complexes (APCs), and transportation stages with a digital twin platform. By digitizing on-site conditions in real time, the system enables managers to intuitively monitor and control the entire distribution

장치의 데이터 무결
보장.